



LAB PROJECT: TRAFFIC LIGHT

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Working Objective:

To simulate real-world traffic light signal control using electronic timing circuits based on 555 ICs in astable mode.

- **Working / Operation :**

- Two NE555 timer ICs act as astable multivibrators (free-running pulse generators).
- Timer-1 controls the Red and Yellow sequence.
- Timer-2 controls the Yellow and Green sequence.
- Both timers generate ON/OFF pulses with different time durations.
- RC combination (resistor + capacitor values) sets timing delays for each LED.
- When output of first 555 goes high → Red LED turns ON.
- After delay, first timer output shifts → Yellow LED switches ON.
- Second 555 output activates → Green LED switches ON while others remain OFF.
- Green switches OFF after its delay → Yellow turns ON again, representing “ready to stop”.
- LEDs glow in realistic traffic order: Red → Yellow → Green → Yellow → Red.
- Output pins drive LEDs via current-limiting resistors to avoid LED damage.
- Both timers run continuously, so sequence repeats indefinitely.
- Circuit behaves like a miniature automatic traffic controller.
- Timing can be altered by changing resistor or capacitor values.
- Powered by a simple 5–9V DC battery, making it portable.

COMPONENTS USED:

Component	Quantity	Purpose
NE555 Timer IC	2	Pulse generation / timing control
Red LED	1	Stop indication
Yellow LED	1	Wait indication
Green LED	1	Go indication
Capacitors 100μF	2	Timing delay
Resistors: 100K, 47K	1 , 1	Timing control
Resistors: 330Ω, 180Ω	2 , 1	LED current limiting
9V Battery	1	Power supply
Breadboard / PCB	1	Circuit mounting
Connecting wires	—	Connections

COMPONENTS DETAILS:

NE555 Timer:

Versatile IC used for timing and pulse generation.

Operates in astable mode here for continuous timing cycle.

LEDs

Converts electrical signals into light; used to represent traffic colours.

Resistors & Capacitors:

R + C sets delay:

$$T = 1.1 \times R \times C$$

Current limiting resistors prevent LED burnout.

Battery (9V):

DC supply to energize the circuit.

Working Flow Chart:

Red LED ON → Delay

Red OFF + Yellow ON → Delay

Green ON → Delay

Green OFF → Yellow ON → Delay

Repeat

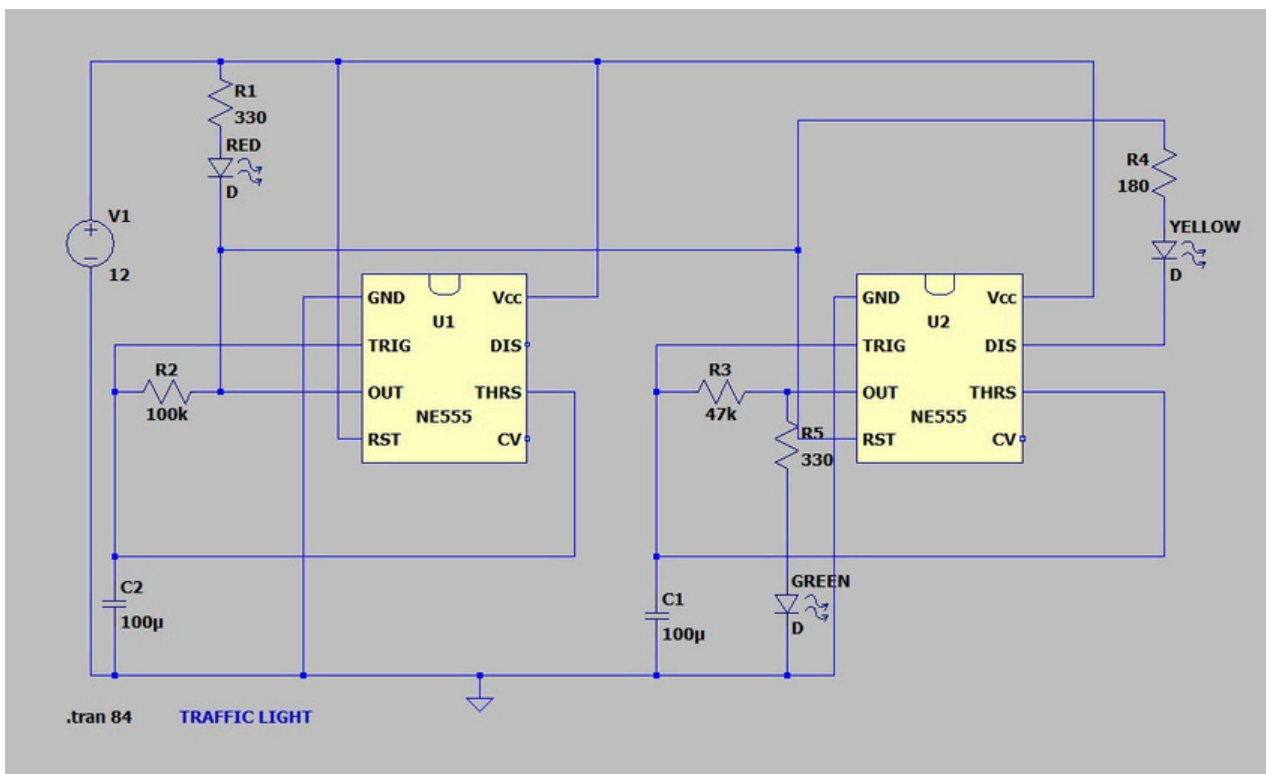
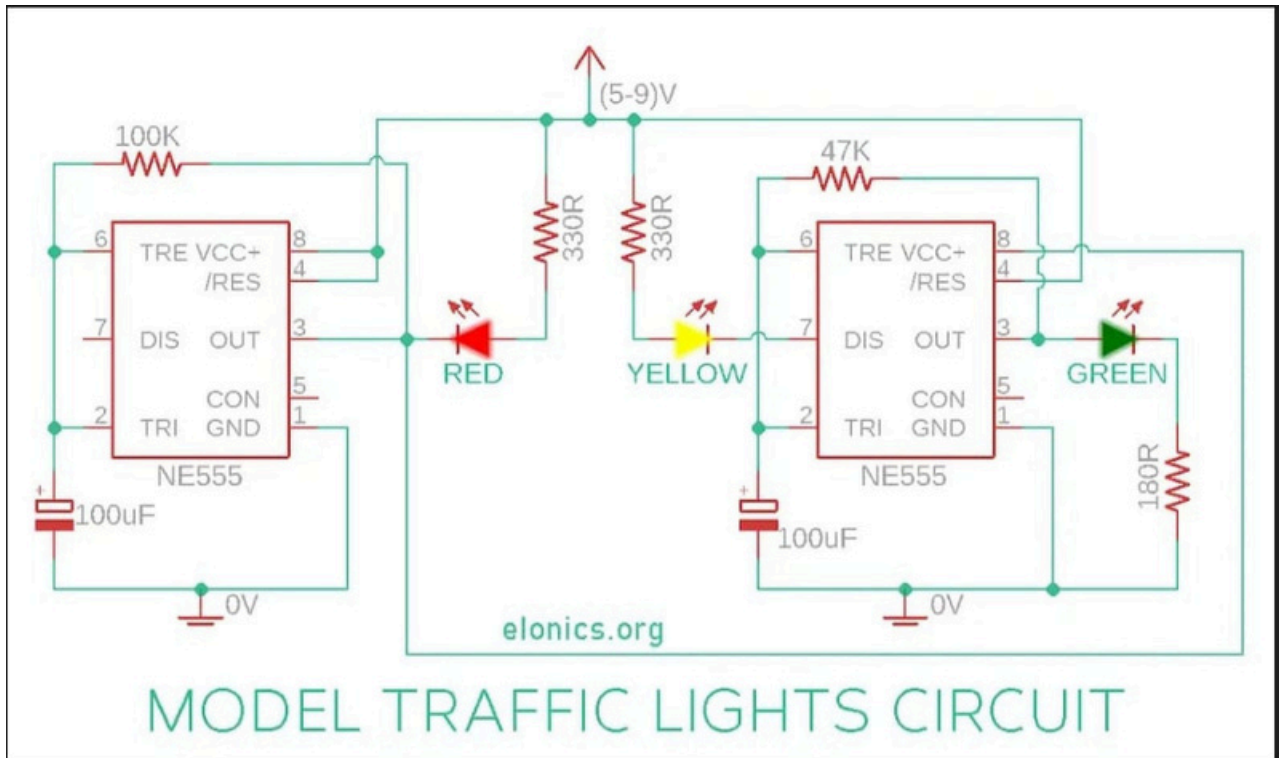
Applications:

Basic traffic automation system model

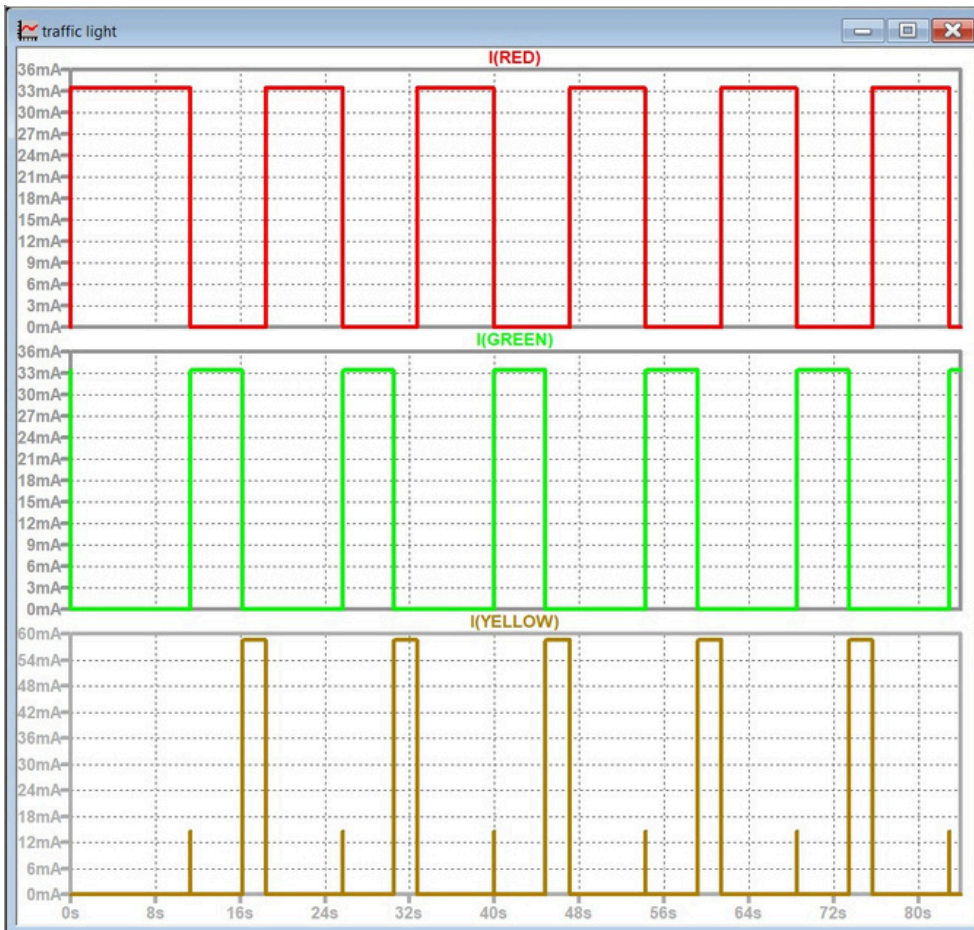
Educational lab demonstration

Real world traffic signal design principle

CIRCUIT DIAGRAM AND LT SPICE SIMULATION:



LT SPICE : SCHEMATIC



RED LIGHT

GREEN LIGHT

YELLOW LIGHT

LT SPICE O/P : WAVEFORM

Bibliography / References

Electronic Circuits for Traffic Light Control – elonics.org

NE555 Datasheet, Texas Instruments

LT-Spice User Manual for Astable Timer Simulation

Arduino / 555 Timing Tutorials – Electronics Hub, CircuitDigest

THANK-YOU